Full Performance Comparison Table

Secure-storage SoC compared with ECG Huffman + CBC-AES paper baseline

Metric	ECG CBC-AES Paper [1]	This Work	Correctness	Improvement (%)
Compression ratio	35.015%	29.87%	**100%**	**+14.69%**
Space saving	64.985%	70.13%	**100%**	**+7.92%**
PRD / byte match	0.411	0 mismatch byte-exact RX	**100%**	**0%**
Compression time	~3.8641 s	1.056 ms TX comp-only	**100%**	**+99.97%**
Decompression time	~0.5818 s	0.483 ms RX Huffman	**100%**	**+99.92%**
Encryption time	~2.7106 s	29.6 us TX AES	**100%**	**+99.999%**
Decryption time	~3.0449 s	69.2 us RX AES	**100%**	**+99.998%**
Compression + encryption	~6.5747 s	1.065 ms TX path	**100%**	**+99.98%**
TX/RX cycles	N/R	53233 TX 24222 RX	**100%**	**0%**
TX/RX throughput	N/R	6.828 MB/s TX-in 15.072 MB/s RX-out	**100%**	**0%**

[1] A lossless compression and encryption mechanism for remote monitoring of ECG data using Huffman coding and CBC-AES, FGCS 2019. Paper platform: MATLAB 2018a on Windows 7 64-bit, i5 2nd Gen, 8 GB RAM. N/R = not reported. RTL timing is measured from performance counters at 50 MHz on five MIT-BIH records.